

# E-Commerce Customer Behavior Analysis using SQL

## Resume Ready SQL Project Documentation

This project focuses on analyzing customer behavior in an e-commerce platform using SQL. The objective is to extract meaningful business insights from customer transactions, product purchases, payment methods, and order history. The project demonstrates advanced SQL concepts such as joins, aggregations, subqueries, window functions, CTEs, ranking functions, and business intelligence reporting.

## Project Objectives

- Analyze customer purchasing behavior.
- Identify top-selling products and categories.
- Track monthly sales performance.
- Measure customer retention and repeat purchases.
- Analyze payment preferences and order trends.
- Generate actionable business insights using SQL.

## Database Schema

| Table Name  | Description                           |
|-------------|---------------------------------------|
| customers   | Stores customer information           |
| products    | Stores product details                |
| orders      | Stores order transaction details      |
| order_items | Stores product-wise order information |
| payments    | Stores payment transactions           |

## SQL Table Creation

```
CREATE TABLE customers (  
  customer_id INT PRIMARY KEY,  
  customer_name VARCHAR(100),  
  gender VARCHAR(10),  
  city VARCHAR(50),  
  signup_date DATE  
);  
  
CREATE TABLE products (  
  product_id INT PRIMARY KEY,  
  product_name VARCHAR(100),  
  category VARCHAR(50),  
  price DECIMAL(10,2)  
);  
  
CREATE TABLE orders (  
  order_id INT PRIMARY KEY,  
  customer_id INT,
```

```

        order_date DATE,
        order_status VARCHAR(30),
        total_amount DECIMAL(10,2),
        FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
    );

CREATE TABLE order_items (
    order_item_id INT PRIMARY KEY,
    order_id INT,
    product_id INT,
    quantity INT,
    FOREIGN KEY (order_id) REFERENCES orders(order_id),
    FOREIGN KEY (product_id) REFERENCES products(product_id)
);

CREATE TABLE payments (
    payment_id INT PRIMARY KEY,
    order_id INT,
    payment_method VARCHAR(30),
    payment_status VARCHAR(30),
    FOREIGN KEY (order_id) REFERENCES orders(order_id)
);

```

## Business Analysis Queries

### 1. Top 10 Customers by Spending

```

SELECT c.customer_name,
       SUM(o.total_amount) AS total_spent
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_name
ORDER BY total_spent DESC
LIMIT 10;

```

### 2. Monthly Revenue Analysis

```

SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
       SUM(total_amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month;

```

### 3. Most Popular Product Categories

```

SELECT p.category,
       SUM(oi.quantity) AS total_quantity_sold
FROM products p
JOIN order_items oi
ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY total_quantity_sold DESC;

```

### 4. Repeat Customers

```

SELECT customer_id,
       COUNT(order_id) AS total_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(order_id) > 1;

```

### 5. Payment Method Analysis

```

SELECT payment_method,
       COUNT(*) AS total_transactions

```

```
FROM payments
GROUP BY payment_method
ORDER BY total_transactions DESC;
```

## 6. Customer Retention Rate

```
WITH repeat_customers AS (
  SELECT customer_id
  FROM orders
  GROUP BY customer_id
  HAVING COUNT(order_id) > 1
)
SELECT
  (COUNT(*) * 100.0 /
   (SELECT COUNT(DISTINCT customer_id) FROM customers))
  AS retention_rate
FROM repeat_customers;
```

## Key Business Insights

- A small percentage of customers contribute to a major portion of total revenue.
- Digital payments are the most preferred payment method among users.
- Certain product categories generate consistently high sales volume.
- Repeat customers significantly increase overall profitability.
- Monthly revenue trends help identify peak sales seasons.

## Skills & Tools Used

| Category      | Skills                                      |
|---------------|---------------------------------------------|
| Database      | MySQL / PostgreSQL                          |
| SQL Concepts  | Joins, CTEs, Window Functions, Aggregations |
| Data Analysis | Customer Segmentation, Revenue Analysis     |
| Reporting     | Business Intelligence Reporting             |

## Resume Description

Developed an SQL-based E-Commerce Customer Behavior Analysis project to analyze customer purchasing trends, monthly revenue, repeat customers, product performance, and payment preferences. Utilized advanced SQL concepts including joins, CTEs, aggregate functions, and business reporting queries to generate actionable business insights.

## Conclusion

This project demonstrates practical SQL skills used in real-world business analytics environments. It showcases the ability to design relational databases, write optimized SQL queries, and generate valuable insights for decision-making in the e-commerce industry.